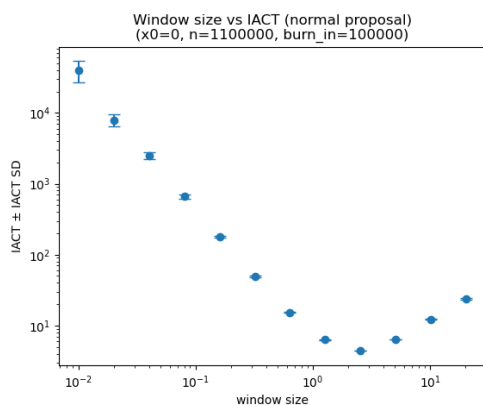


## Compute 2

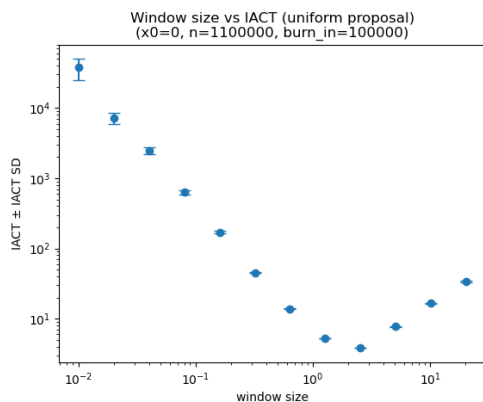
### Exercise 1

(a)



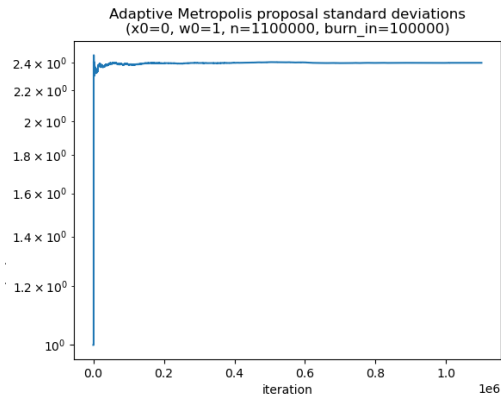
The optimal window size seems to be about  $w = 2.56$  which gives an IACT estimate of 4.4190 with a standard deviation of 0.0346.

(b)



The optimal window size again seems to be about  $w = 2.56$  which gives an IACT estimate of 3.8309 with a standard deviation of 0.0282. (Here, 'window size' means the proposal is  $wU(-\sqrt{12}/2, \sqrt{12}/2)$ , so that the standard deviation of the proposal is  $w$ .)

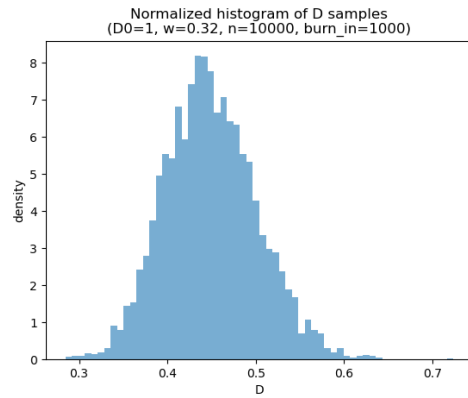
(c)



The optimal window size again seems to be about  $w = 2.40$  which gives an IACT estimate of 4.3737. with a standard deviation of 0.0342. Yes, this does achieve the optimum found in part (a).

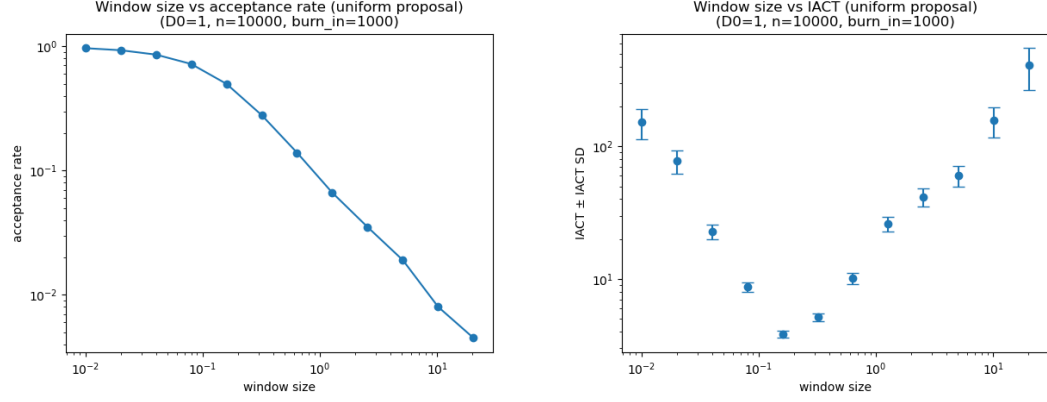
## Exercise 2

(a)



The mean is 0.449 and the standard deviation is 0.0524.

(b)

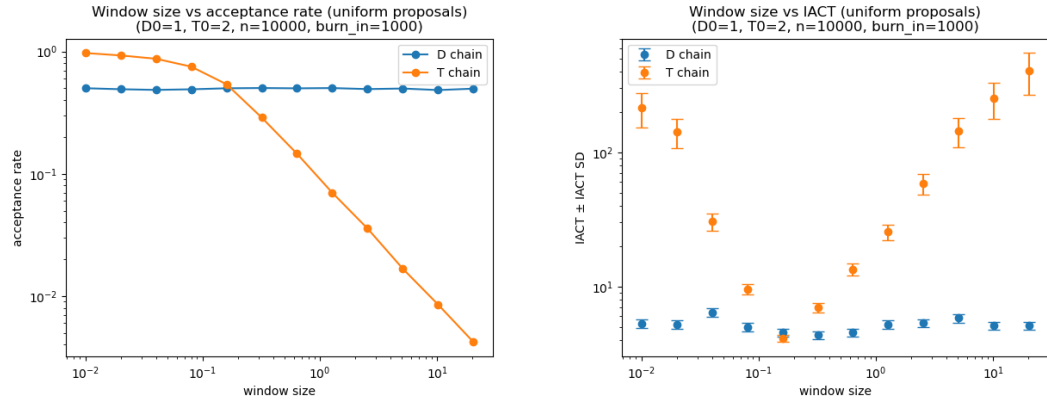


The optimal proposal window size is the value that gives to the lowest IACT. In our case, this optimal value seems to be about  $w = 0.16$  which gives the lowest IACT estimate of 3.8258 with a standard deviation of 0.2363. The corresponding acceptance rate is 0.49 which seems reasonable.

(c)

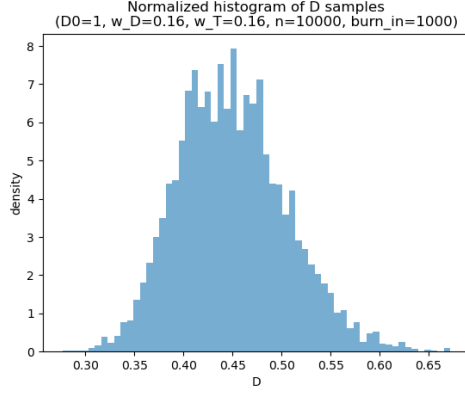
I have used a Metropolis-within-Gibbs sampler with uniform proposals for D and T. When initially running the sampler with proposal window  $w_D = 0.16$  for D (which was identified as optimal in part (b)), and proposal window  $w_T = 0.03$  for T (which was a wild guess), the trace plot for T was far from a fuzzy caterpillar.

As such, to identify the optimal proposal window  $w_T$  for T, I have run the sampler for various values of  $w_T$  to produce the following plots:



(I think) that since we ultimately would like to perform joint inference over D (and T is a nuisance parameter), the optimal value of  $w_T$  is the value that gives the lowest IACT of the D chain. However, the IACT of the D chain seems to

stay roughly constant. So, I will choose  $w_T = 0.16$  which gives the lowest IACT estimate for the T chain.



The mean is 0.450 and the standard deviation is 0.0552.

We indeed observe that uncertainty in the value of  $T$  gives us more uncertainty in the value of  $D$  (the standard deviation is now slightly larger). However, both densities look very similar.

### Exercise 3

Consider the joint pdf  $\pi(x, y, \theta)$  over  $x$ ,  $y$ , and  $\theta$ . Using Bayes' rule, product factoring, etc., establish the following identities between the pdfs over various conditional and marginal distributions:

(a)

$$\pi(\theta | y) = \frac{\pi(x, \theta | y)}{\pi(x | \theta, y)}.$$

(This is useful because the RHS may be evaluated at *any*  $x$  for which the denominator is non-zero.)

*Proof.* Fix  $\theta$  and  $y$  so that  $\pi(y), \pi(\theta, y) \neq 0$ . Fix  $x$  so that  $\pi(x | \theta, y) \neq 0$ . By the definition of conditional density

$$\pi(x | \theta, y) \pi(\theta | y) = \frac{\pi(x, \theta, y)}{\pi(\theta, y)} \frac{\pi(\theta, y)}{\pi(y)} = \frac{\pi(x, \theta, y)}{\pi(y)} = \pi(x, \theta | y).$$

Rearranging gives the result.  $\square$

(b)

$$\pi(x, \theta | y) = \pi(x | \theta, y) \pi(\theta | y),$$

i.e., we can condition the usual product factoring.

*Proof.* Fix  $\theta$  and  $y$  so that  $\pi(y), \pi(\theta, y) \neq 0$ . Fix  $x$ . If  $x$  satisfies  $\pi(x | \theta, y) \neq 0$ , the result immediately follows from (a).

Otherwise, if  $x$  satisfies  $\pi(x | \theta, y) = 0$ , by the definition of conditional density we have that

$$0 = \pi(x | \theta, y) = \frac{\pi(x, \theta, y)}{\pi(\theta, y)},$$

which implies that  $\pi(x, \theta, y) = 0$ . Again by the definition of conditional density, this implies that  $\pi(x, \theta | y) = 0$ . Thus, in this case the identity also holds.  $\square$

(c)

$$\pi(\theta | y) = \frac{\pi(x, y | \theta) \pi(\theta)}{\pi(x | \theta, y) \pi(y)}.$$

(Again, RHS may be evaluated at *any*  $x$  for which denominator is non-zero. Useful when full conditional over  $\theta$  is a GMRF.)

*Proof.* Fix  $\theta$  and  $y$  so that  $\pi(\theta), \pi(y), \pi(\theta, y) \neq 0$ . Fix  $x$  so that  $\pi(x | \theta, y) \neq 0$ . From part (a), the definition of conditional density, and the product rule we have that

$$\pi(\theta | y) = \frac{\pi(x, \theta | y)}{\pi(x | \theta, y)} = \frac{\pi(x, \theta, y)}{\pi(x | \theta, y) \pi(y)} = \frac{\pi(x, y | \theta) \pi(\theta)}{\pi(x | \theta, y) \pi(y)}.$$

$\square$

## Exercise 4

Assume the  $x_1, x_2, \dots$  are the sequence of states generated by a Markov process (perhaps an MCMC) that is ergodic with respect to some distribution  $\pi(x)$ . Let  $y = g(x)$  for some function  $g$  of  $x$ , with corresponding pdf  $\pi(y)$  (given by the change of variable formula).

(a)

Show that  $g(x_1), g(x_2), \dots$  are the sequence of states generated by a Markov process that is ergodic with respect to  $\pi(y)$ .

Let  $(X_n)_{n \in \mathbb{N}}$  be a Markov chain on state space  $\mathcal{X}$  which is ergodic wrt  $\pi_X$ . That is,

$$\lim_{n \rightarrow \infty} \|K_X^n(x, \cdot) - \pi_X\|_{TV} = \lim_{n \rightarrow \infty} \sup_{A \in \mathcal{B}(\mathcal{X})} |K_X^n(x, A) - \pi_X(A)| = 0$$

for all  $x \in \mathcal{X}$ , where  $K_X$  is the transition kernel of the chain  $(X_n)_{n \in \mathbb{N}}$ .

Assume  $g : \mathcal{X} \rightarrow g(\mathcal{X})$  is a measurable bijection with measurable inverse. Let  $(Y_n)_{n \in \mathbb{N}} := (g(X_n))_{n \in \mathbb{N}}$  be a chain on state space  $\mathcal{Y} := g(\mathcal{X})$ , and let  $\pi_Y = \pi_X \circ g^{-1}$  be the distribution of  $Y := g(X)$  where  $X \sim \pi_X$ . We will show that

- i.  $(Y_n)_{n \in \mathbb{N}}$  is a Markov chain with transition kernel  $K_Y : \mathcal{Y} \times g(\mathcal{Y}) \rightarrow [0, 1]$  defined by  $K_Y(y, A) := K_X(g^{-1}(y), g^{-1}(A))$ .
- ii.  $(Y_n)_{n \in \mathbb{N}}$  is ergodic wrt  $\pi_Y$ .

*Proof.*

- i. Fix  $n \in \mathbb{N}$  and  $A \in \mathcal{B}(\mathcal{Y})$ . We aim to show that

$$P(Y_{n+1} \in A | Y_0, \dots, Y_n) = P(Y_{n+1} \in A | Y_n) = K_Y(y_n, A).$$

To this end, since  $g$  is a measurable bijection with measurable inverse, and  $Y_m = g(X_m)$ , we have  $\sigma(X_m) = \sigma(Y_m)$  for each  $m$ . This implies that  $P(B | X_m) := E[1_B | \sigma(X_m)] = E[1_B | \sigma(Y_m)] =: P(B | Y_m)$  for any event  $B$ . It then follows that

$$\begin{aligned} P(Y_{n+1} \in A | Y_0, \dots, Y_n) &= P(Y_{n+1} \in A | X_0, \dots, X_n) \\ &= P(g(X_{n+1}) \in A | X_0, \dots, X_n) \\ &= P(X_{n+1} \in g^{-1}(A) | X_0, \dots, X_n) \\ &= P(X_{n+1} \in g^{-1}(A) | X_n) \quad ((X_n)_{n \in \mathbb{N}} \text{ is a Markov chain}) \\ &= P(g(X_{n+1}) \in A | X_n) \\ &= P(Y_{n+1} \in A | X_n) \\ &= P(Y_{n+1} \in A | Y_n). \end{aligned}$$

That is,

$$P(Y_{n+1} \in A | Y_0, \dots, Y_n) = P(Y_{n+1} \in A | Y_n).$$

Moreover, since  $K_X$  is the transition kernel of  $(X_n)_{n \in \mathbb{N}}$ , we have

$$P(Y_{n+1} \in A | Y_n) = P(X_{n+1} \in g^{-1}(A) | X_n) = K_X(X_n, g^{-1}(A)) = K_X(g^{-1}(Y_n), g^{-1}(A)) = K_Y(Y_n, A).$$

So, overall,

$$P(Y_{n+1} \in A | Y_0, \dots, Y_n) = P(Y_{n+1} \in A | Y_n) = K_Y(y_n, A).$$

Hence,  $(Y_n)_{n \in \mathbb{N}}$  is a Markov chain with transition kernel  $K_Y$ .

- ii. Fix  $y \in \mathcal{Y}$ . We aim to show that

$$\lim_{n \rightarrow \infty} \|K_Y^n(y, \cdot) - \pi_Y\|_{TV} = \lim_{n \rightarrow \infty} \sup_{A \in \mathcal{B}(\mathcal{Y})} |K_Y^n(y, A) - \pi_Y(A)| = 0.$$

To this end, let  $x = g^{-1}(y)$ . Since  $(X_n)_{n \in \mathbb{N}}$  is ergodic wrt  $\pi_X$ , we have

$$\begin{aligned} 0 &= \lim_{n \rightarrow \infty} \|K_X^n(x, \cdot) - \pi_X\|_{TV} \\ &= \lim_{n \rightarrow \infty} \sup_{A \in \mathcal{B}(\mathcal{X})} |K_X^n(x, A) - \pi_X(A)| \\ &= \lim_{n \rightarrow \infty} \sup_{A \in \mathcal{B}(\mathcal{X})} |K_Y^n(g(x), g(A)) - \pi_X(A)| \quad (\text{by the definition of } K_Y) \\ &= \lim_{n \rightarrow \infty} \sup_{A \in \mathcal{B}(\mathcal{X})} |K_Y^n(g(x), g(A)) - \pi_X \circ g^{-1}(g(A))| \\ &= \lim_{n \rightarrow \infty} \sup_{A \in \mathcal{B}(\mathcal{X})} |K_Y^n(y, g(A)) - \pi_Y(g(A))|. \end{aligned}$$

One can show that since  $g$  is measurable and bijective with measurable inverse,

$$\{g(A) : A \in \mathcal{B}(\mathcal{X})\} = \{A : A \in \mathcal{B}(g(\mathcal{X}))\}.$$

Using this, we have

$$\lim_{n \rightarrow \infty} \sup_{A \in \mathcal{B}(\mathcal{Y})} |K_Y^n(y, A) - \pi_Y(A)| = \lim_{n \rightarrow \infty} \|K_Y^n(y, \cdot) - \pi_Y\|_{TV} = 0.$$

Therefore,  $(Y_n)_{n \in \mathbb{N}}$  is ergodic wrt  $\pi_Y$ .

□

(b)

Show that if the  $n$ -step distribution over  $x_i$  converges in distribution, then so does the  $n$ -step distribution over  $y_i$ .

*Proof.* Assume that  $g$  is continuous. Suppose that  $X_n$  converges in distribution to  $X$ . That is,

$$\lim_{n \rightarrow \infty} E[f(X_n)] = E[f(X)]$$

for all bounded and continuous functions  $f : \mathcal{X} \rightarrow \mathbb{R}$ . We aim to show that  $Y_n$  converges in distribution to  $Y$ . To this end, let  $h : \mathcal{Y} \rightarrow \mathbb{R}$  be bounded and continuous. Observe that for each  $n$

$$E[h(Y_n)] = E[h(g(X_n))] = E[(h \circ g)(X_n)].$$

Since  $g$  is continuous,  $h \circ g$  is continuous and bounded. As  $X_n$  converges in distribution to  $X$  it follows that

$$\lim_{n \rightarrow \infty} E[h(Y_n)] = \lim_{n \rightarrow \infty} E[(h \circ g)(X_n)] = E[(h \circ g)(X)] = E[h(Y)].$$

□

**Let me know if I used definitions / made assumptions you did not intend here.**